Free-body diagrams

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Draw only the external forces on one chosen object.

SUCCESS CRITERIA - I can...

1. Choose one object and list its external interactions.
2. Draw force arrows in physically correct directions.
3. Remove velocity and net force as extra interaction arrows.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal force” and “tension” to explain your model or result. Sentence starters are optional.

normal force: a support force perpendicular to a contact surface

tension: the pulling force through a stretched rope or cable

weight: the gravitational force on an object

01 Bring back an earlier idea

An a-versus-F_{net} graph has slope 1.5 $(m/s^2)/N$.
Find mass and acceleration at 2.5 N.
Explain why a driving-force plot can have an offset.

Word help: mass = a measure of how strongly an object resists acceleration

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Sigma F_x = ma_x$; $\Sigma F_y = ma_y$; $F_g = mg$.
Draw only forces on the chosen object. x and y name directions. Use g = 9.8 N/kg unless another value is given.

Worked reasoning: A lamp hangs motionless from one vertical cable. With weight 12 N down, the cable tension is 12 N up. Only forces acting on the lamp belong in its diagram.

Your turn: Draw the lamp as a labeled point with its two force arrows.
Explain why its force on the cable is not another arrow on this diagram.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Motionless lamp

A 2 kg lamp hangs from one vertical rope.
Draw and label the free-body diagram, including magnitudes.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Add and label force arrows.

The dot represents the object. Object: 2 kg lamp

Marzano check - circle: ① ② ③

04 A rough ramp

A block rests on a rough incline.
Draw weight, normal and static friction. Orient each arrow correctly; explain why weight is not perpendicular to the ramp.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Critique the extra arrows

A diagram of a sliding box includes weight, normal, friction,“velocity force”and“net force.”What must be removed or redrawn?
Explain.

Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

06 Choose a method

A straight position-time segment goes from (2 s, -4 m) to (6 s, 20 m).
Find velocity, interpret its units and predict x at 8 s if motion continues.

Word help: velocity = rate of position change, with direction

Sentence starter: I predict … . My model supports this because … .

Marzano check - circle: 1 2 3

07 Exit check

A box slides to the left on a rough level floor after release.
Draw all forces; state what can be concluded about vertical and horizontal acceleration.

Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 4 kg object hangs from two symmetric cables, each 30° from vertical.
Use g=10 N/kg.
Draw the diagram and calculate each tension.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3